



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

ENS221: Environmental Science

Mid-term Examination

Semester: Fall 2024

Time: 1 Hour 30 Minutes

Total Marks: 30

Date: 06/01/2025

Part A (Answer any Two)

		Marks	CLO
1.	a) Define Environmental Science. Differences between Environmental Science and Environmental Studies.	5	CLO1
	b) 'Environmental Science is a multi-disciplinary approach'. Explain.	5	CLO1
2.	a) Give a description of eutrophication in your own words.	4	CLO1
	b) Develop an argument between developed country and developing country regarding greenhouse gases emission.	6	CLO1
3.	a) What do you understand by pollutant? Sketch point-source and non-point source of pollutant.	3	CLO1
	b) Global warming is a part of climate change', Justify it.	2.5	CLO1
	c) Why do most weather phenomena, like clouds and storms, occur in the troposphere? 'The upper part of the thermosphere is known as ionosphere'. Explain.	4.5	CLO1

Part B (Answer any One)

4.	a) What do you think about the main idea of 'Sustainable Development'? Write the historical background (1957-2015) of it.	5	CLO2
	b) 'The three zero theory will help to reduce the greenhouse gasses emission'. Clarify it.	5	CLO2
5.	a) What do you understand by food security?	2	CLO2
	b) Apply sustainable agriculture and climate smart agriculture for food security.	3	CLO2
	c) Bangladesh's vast low-lying floodplain makes it very vulnerable to increasing sea levels. Examine the possible effects of rising sea levels on Bangladesh's coastal areas.	5	CLO2

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 227: Data Communication

Batch: 24th

Date: 08/01/2025

Mid-Term Examination

Semester: Fall 2024

Time: 1 Hour 30 Minutes

Total Marks : 30

Answer any five questions:

5 × 6 = 30

1. (a) Explain the five components of a data communication system. CLO1 3
(b) What are the differences between simplex and half-duplex transmission? CLO1 3
2. (a) Write the differences between the duties of the IETF and IRTF. CLO1 3
(b) For 10 devices in a network, calculate the number of cable links required for a Mesh, Ring, Bus and Star topology. CLO1 3
3. (a) Briefly describe the layers of TCP/IP protocol suite. CLO4 3
(b) Assume we have created a packet-switched network and use the TCP/IP protocol suite. CLO4 3
What are the advantages and disadvantages of sending large packets using this network?
4. (a) Explain Multiplexing and De-multiplexing in TCP/IP protocol suite. CLO4 3
(b) Write the phase shift for the following conditions: CLO3 3
 - i. A sine wave with the maximum amplitude at time zero
 - ii. A sine wave with maximum amplitude after 1/4 cycle
 - iii. A sine wave with zero amplitude after 3/4 cycle and increasing
5. (a) Describe Nyquist Bit Rate and Shannon Capacity. CLO3 3
(b) Name the three types of transmission impairment. Explain. CLO5 3
6. (a) List and explain five line coding schemes. CLO2 3
(b) Distinguish between signal element and data element. CLO3 3
7. Write short notes on: CLO5 3×2=6
 - i) Baseline Wandering.
 - ii) DC Components.
 - iii) Self-synchronization.

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE225: Digital Logic Design

Batch: 24th

Date: 14/01/2025

Midterm Examination

Semester: Fall 2024

Time: 1 Hour 30 minutes

Total Marks: 30

Answer any **five** questions of the following.

$5 \times 6 = 30$

1. (a) Find the 1's and 2's complements of the following binary numbers: CLO2 3
 i. $(1011.101)_2$ ii. $(1101.011)_2$
- (b) A 4-bit data word uses even parity for error detection. How many parity bits are needed? If the data word is 1101, what will be the complete transmitted word? CLO1 3
2. (a) Perform the following binary subtraction using 1's complement method: $(1000100)_2 - (1010100)_2$ CLO2 3
- (b) Using 9's complement method, solve: $(28321)_{10} - (76425)_{10}$ CLO2 3
3. (a) Reduce the following Boolean expression using Boolean algebra laws: $F(A,B,C,D) = AB'C'D + ABC'D + AB'CD' + ABCD' + A'BCD + ABC'D' + A'B'CD$ CLO3 3
- (b) Simplify the following Boolean expression up to 4 literals: $\bar{A}\bar{C} + \bar{A}\bar{C} + ABC$ CLO3 3
4. (a) Convert the following Boolean expression to Canonical SOP form using K-map: $(AB + C)(B + A'B'C)$ CLO2 2
- (b) $F(w,x,y,z) = \sum m(1,4,5,6,7,9,11,13,15) + d(0,2,8)$ CLO4 4
 i. Find both minimal SOP and POS expressions.
 ii. Compare the cost.
 iii. Draw the logic diagrams.
 iv. Determine which form is more efficient for this function.
5. (a) Convert the following Boolean expression into POS: $X' + X(X + Y')(Y + Z')$ CLO2 2
- (b) Design a circuit that converts excess-3 code to BCD. Use K-maps for minimization and implement using only NAND gates. CLO4 4

$A B + A$

6. (a) What are the differences between Minterm and Implicant? Explain with proper example. CLO3 2
- (b) Design a 4-bit binary to gray code converter using K-map minimization. CLO4 4
7. (a) Create a pattern detector circuit that: CLO3 3
- i. Monitors a 4-bit sequence
 - ii. Outputs 1 when it detects any 3 consecutive 1's
(Use K-map to minimize the logic)
- (b) $F(w,x,y,z) = \sum m(0, 1, 4, 6, 7, 9, 13, 14) + d(3, 5, 8, 11)$ CLO3 3
- i. Find all prime implicants
 - ii. Identify essential prime implicants



CSE221: Computer Algorithms

Mid-Term Examination

Semester: Fall 2024

Time: 1 Hour 30 Minutes

Total Marks: 30

Batch: 24th
Date: 01-01-2025

- | | CLOs | Marks | | | | | | | | | | | | | | | | | | | | | | | | |
|--|---------|-------|----|----|----|-----|----|----|---|----|----|----|----------|---|----|----|----|----|-----|----|----|---|----|----|------|---|
| 1. (a) Explain the criteria of an algorithm. | CLO1 | 2 | | | | | | | | | | | | | | | | | | | | | | | | |
| (b) Define asymptotic notation. Describe three asymptotic notations used to compute the efficiency of an algorithm. | CLO1 | 4 | | | | | | | | | | | | | | | | | | | | | | | | |
| 2. (a) Write the properties of Red-Black Tree. | CLO3 | 2 | | | | | | | | | | | | | | | | | | | | | | | | |
| (b) Analyze the complexity from the following recurrence:
$T(n) = \begin{cases} 1 & \text{if } n \leq 1 \\ 2 \times T\left(\frac{n}{2}\right) + n & \text{otherwise} \end{cases}$ | CLO1 | 4 | | | | | | | | | | | | | | | | | | | | | | | | |
| 3. (a) Define feasible solution and optimal solution. | CLO3 | 2 | | | | | | | | | | | | | | | | | | | | | | | | |
| (b) Construct the 'Merge' function to use in Merge Sort Algorithm. | CLO5 | 4 | | | | | | | | | | | | | | | | | | | | | | | | |
| 4. (a) Find the location of 45 in the following array using Interpolation Search:
IT[8]: 35, 40, 45, 50, 55, 60, 65, 70 | CLO5 | 2 | | | | | | | | | | | | | | | | | | | | | | | | |
| (b) Suppose you have 11 elements in an array as follows:
<table border="1"> <thead> <tr> <th>Address</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> </tr> </thead> <tbody> <tr> <td>Elements</td> <td>5</td> <td>-8</td> <td>35</td> <td>11</td> <td>15</td> <td>-17</td> <td>25</td> <td>31</td> <td>2</td> <td>12</td> <td>-1</td> </tr> </tbody> </table> Draw the recursion tree to find the maximum and minimum elements from the array using recursive Mini-Max Algorithm. | Address | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | Elements | 5 | -8 | 35 | 11 | 15 | -17 | 25 | 31 | 2 | 12 | -1 | CLO2 | 4 |
| Address | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | | | | | | | | | | | | | | | |
| Elements | 5 | -8 | 35 | 11 | 15 | -17 | 25 | 31 | 2 | 12 | -1 | | | | | | | | | | | | | | | |
| 5. (a) Explain the components of Greedy Method. | CLO5 | 2 | | | | | | | | | | | | | | | | | | | | | | | | |
| (b) Let you have four slots and six jobs with the following profits and deadlines respectively:
$\{p_1, p_2, p_3, p_4, p_5, p_6\} = \{20, 17, 12, 10, 9, 8\}$ and
$\{d_1, d_2, d_3, d_4, d_5, d_6\} = \{3, 1, 4, 1, 3, 2\}$
If each job takes one unit of time to execute, find the optimal solution using Greedy Algorithm. | CLO5 | 4 | | | | | | | | | | | | | | | | | | | | | | | | |
| 6. (a) Assume you have three files with the length 8, 12 and 7 respectively. Find the optimal ordering of the files for retrieving if the storing device is sequential. | CLO5 | 3 | | | | | | | | | | | | | | | | | | | | | | | | |
| (b) Consider the following instance of knapsack problem:
Number of objects, $n = 6$;
Capacity of knapsack, $m = 14$;
Profits, $\{p_1, p_2, p_3, p_4, p_5, p_6\} = \{10, 5, 15, 17, 21, 18\}$ and
Weights, $\{w_1, w_2, w_3, w_4, w_5, w_6\} = \{3, 2, 4, 4, 4, 5\}$.
Calculate the optimal solution using Greedy Method. | CLO5 | 3 | | | | | | | | | | | | | | | | | | | | | | | | |



In Pursuit of Knowledge

HAMDARD UNIVERSITY BANGLADESH

7. (a) How does the use of recursion affect an algorithm's time and space complexity? CLO2 2
- (b) Let, you have two strings A= "OFM" and B= "BOMT". Now calculate the longest common subsequence using recursion. CLO2 4

Faculty of Science, Engineering and Technology
 Department of Computer Science and Engineering

CSE 229 : Computer Architecture

Batch: 24th

Semester: Fall 2024

Date: 12/01/2025

Midterm Examination

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions.

5 × 6 = 30

1. (a) Define Computer Architecture and Computer Organization. CLO1 2
 (b) Illustrate a top-down view of a Computer Structure. State the distinction between Structure and Function. CLO1 4
2. (a) What is Hardware Description Language? CLO1 2
 (b) Explain the differences between structure and behavior in the computer architecture. Illustrate your answer by giving a purely structural and behavioral description of an EXCLUSIVE-OR circuit. CLO1 4
3. (a) Mention the three levels of abstraction in the design of a computer system. CLO1 2
 (b) Illustrate the logic circuit of a serial adder and construct a state table stating its behavior. CLO1 4
4. (a) What is the design problem for register-level systems? CLO1 2
 (b) Draw a register-level system containing a multifunction ALU that can perform multiple operations. CLO1 4
5. (a) Illustrate how the ALU is interconnected with the rest of the processor by indicating its inputs and outputs. CLO2 2
 (b) Draw the flowchart for the unsigned binary multiplication algorithm using the shift-and-add method. CLO3 4
6. (a) What is Range Extension in number representation? How is Range Extension accomplished in sign-magnitude notation? CLO1 2
 (b) Perform the binary shift-and-add method to multiply the 4-bit unsigned binary numbers $(101)_2$ (Multiplicand) and $(1101)_2$ (Multiplier). Show the change in values of the carry (C), accumulator (A), and multiplier (Q) after each cycle. CLO3 4
7. (a) State the Overflow rule in the ALU. CLO1 2
 (b) Draw the block diagram of an arithmetic unit that uses a complementer and a control switch to perform binary addition and subtraction with overflow detection. CLO2 4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 223: Object Oriented Programming

Batch: 24th

Semester: Fall 2024

Time: 1 Hour 30 Minutes

Date: 04/01/2025

Mid-Term Examination

Total Marks: 30

Answer five questions of the followings.

5 × 6 = 30

1(a). Briefly explain the functionalities of Java Virtual Machine (JVM).

CLOs Marks

CLO1 2

1(b). You need to write a method `editStudentDetails()` that takes name, age, and CGPA as input. You will modify these values inside the method, and these changed values should be available outside the method. Show in the `main()` body of the program how do you call this method and access those changed values.

CLO2 4

2(a). Briefly explain any three features that differentiate Object-Oriented Programming from Structured Programming Languages.

CLO1 2

2(b). Write a class named `TwoDimMatrix` that operates on a two dimension `m x n` matrix as input. You need to write necessary constructors and following methods: `input()` to take input of a matrix, `print()` to display the matrix, `determinant()` to find the determinant, and `add(A)` to addition the matrix by another matrix `B`.

CLO2 4

3(a). Write the following code using `for` loop and show the output.

CLO3 3

```
int i=1;
do{
    int j=1;
    do{
        System.out.print(i);
        j++;
    }while(j<=i);
    System.out.println();
    i++;
}while(i<=5);
```


- 3(b). Write a Java program for the following series- CLO2 3
- $$1^2 - 2^2 + 3^2 - 4^2 - \dots +/ - n^2$$
- 02 1
- 4(a). What is a **constructor** in Java? What is the prime difference between constructor and method? CLO3 2
- 4(b). Write a program in Java to find the maximum number between two numbers and among three numbers using **method overloading**. CLO2 4
- 5(a). Write a program in Java/C++ for **Sequential Search** of an array. CLO2 2
- 5(b). An extended stack can perform three operations: **push**, **pop** and **append**. The push operation pushes an element into top of the stack; the pop operation extracts the top element from the stack and the append operation adds or concatenates the given element with the top element of the stack. Write down a generic class for implementing the extended stack that can act as the container for the data type of integer, floating, double and String. CLO2 4
- 6(a). Write about '**extends**' keyword in Object Oriented Programming. CLO3 2
- 6(b). Write a method that takes an array of integers as an argument and returns a value based on the sums of the even and odd numbers in the array. CLO2 4
- Let X = the sum of the odd numbers in the array and let Y = the sum of the even numbers. The method should return X - Y.
- 7(a). Briefly explain '**final variable**' in Object Oriented Programming with example. CLO3 2
- 7(b). An array with an odd number of elements is said to be centered if all elements (**except the middle one**) are strictly greater than the value of the middle element. Note that only arrays with an odd number of elements have a middle element. Write a method that accepts an integer array and returns 1 if it is a centered array, otherwise it returns 0. CLO2 4